

The Sato–Tate inputs for a Maass form: an exact reduction, the moment record, and structural insufficiency theorems

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Abstract

Let π be a Hecke–Maass cusp form for $\mathrm{SL}_2(\mathbb{Z})$ with Satake parameters α_p and angles θ_p . The Sato–Tate conjecture for π — temperedness at every prime together with equidistribution of $\{\theta_p\}$ for the semicircle measure — remains open; unlike the holomorphic case it is untouched by symmetric-power modularity of Newton–Thorne, whose method does not reach Maass forms. This note records, with no conditional statement anywhere except where a hypothesis is explicitly named as such: (1) an *exact* two-input reduction of the conjecture, each input equivalent to the conclusion it feeds, so that the interface provably has zero slack; (2) the unconditional moment record — the first nine moments of $\{\theta_p\}$ converge to the Sato–Tate moments (Kim–Shahidi), with the complete tensor-tower coefficient algebra behind the record machine-checked, and the local Rankin–Selberg factor exhibited as an exact sum of squares; (3) an extension of the reachable *joint* moment matrix for a pair of Maass forms by the three entries $(5, 1)$, $(6, 1)$, $(5, 2)$, via automorphic vehicle identities, stated at their precise analytic grade; (4) two structural insufficiency theorems delimiting what any future proof must contain: no Satake-independent analytic payload can supply converse-theorem inputs, and every factorization of the missing $\mathrm{GL}(4)$ -twist family for Sym^5 through the monoid of known automorphic objects requires one of exactly two unknown functorial products; and (5) the frontier stated exactly: Sym^5 automorphy — the gateway past the record — has two equivalent faces, one new functorial product or a global converse theorem at half twist range together with converse-grade entirety, and beyond finitely many rungs any route to the full conjecture requires an engine not present in current mathematics.

1 The conjecture and the exact reduction

Fix a Hecke–Maass cusp form π for $\mathrm{SL}_2(\mathbb{Z})$, unramified everywhere, with unitarized Satake parameters $\{\alpha_p, \alpha_p^{-1}\}$ and $S_r(\alpha) = \sum_{j=0}^r \alpha^{r-2j}$ the symmetric-power trace ladder. The Sato–Tate conjecture asserts (i) $|\alpha_p| = 1$ for every p , and (ii) equidistribution of the angles θ_p with respect to $\frac{2}{\pi} \sin^2 \theta d\theta$.

Theorem 1.1 (exact reduction; machine-checked). *Let M be the seed datum of π . Then:*

*Machine-checked artifacts cited in the text live in the `RequestProject` Lean corpus of the helix–Frobenius repository (named files per section); every cited declaration compiles with axiom footprint `{propext, Classical.choice, Quot.sound}`, no sorry, no axiom. sam@sk-labs.com

1. a per-rung polynomial ceiling $\|\alpha_p^{\pm r}\| \leq C_p(r+1)^k$ for all $r \geq 1$ exists if and only if π is tempered at every prime;
2. given temperedness, the vanishing of the prime averages of $S_r(\alpha_p)$ for every $r \geq 1$ is equivalent to Sato–Tate equidistribution of the angles.

Consequently any proof of the conjecture must produce these two inputs at full strength: the interface has zero slack.

The two equivalences are the compiled statements `temperedness_iff_polyCeiling` and `cancellation_iff_equidistribution`, assembled in `maass_satoTate_exact` (file `MaassSatoTateClosure`); the polynomial-to-uniform ceiling manufacture is `MaassPolyCeiling.toCeiling`. Serre’s criterion [15] converts input (2) into the one-line theory of $L(\text{Sym}^r \pi, s)$: holomorphy and nonvanishing on $\text{Re } s \geq 1$, for every r .

2 The unconditional moment record

Theorem 2.1 (Kim–Shahidi; Gelbart–Jacquet, Kim). *For π as above and generic (non-dihedral) type: $\text{Sym}^r \pi$ is automorphic for $r \leq 4$ [1, 3, 4]; $L(\text{Sym}^r \pi, s)$ admits meromorphic continuation with functional equation for $r \leq 9$, holomorphic and nonvanishing on $\text{Re } s \geq 1$ in the range needed by Serre’s criterion [5, 6]. Hence the first nine moments of $\{\theta_p\}$ converge to the Sato–Tate moments.*

Three layers behind this record are machine-checked in the corpus:

Proposition 2.2 (tensor-tower coefficient algebra; machine-checked). *For a Hecke eigen-datum with Satake α , the coefficient banks satisfy, as exact identities of arithmetic functions at every n : $\text{std} \otimes \text{Sym}^2 = \text{Sym}^3 \oplus \text{std}$, $\text{std} \otimes \text{Sym}^4 = \text{Sym}^5 \oplus \text{Sym}^3$, $\text{Sym}^2 \otimes \text{Sym}^4 = \text{Sym}^6 \oplus \text{Sym}^4 \oplus \text{Sym}^2$, $\text{Sym}^3 \otimes \text{Sym}^4 = \text{Sym}^7 \oplus \text{Sym}^5 \oplus \text{Sym}^3 \oplus \text{std}$, $\text{Sym}^4 \otimes \text{Sym}^4 = \text{Sym}^8 \oplus \text{Sym}^6 \oplus \text{Sym}^4 \oplus \text{Sym}^2 \oplus 1$, together with their L -series forms on the common half-plane of absolute convergence.*

(Files `TensorTowerBanks`, declarations `tensorCube_bank` through `tensorSym4Sym4_bank` and the `LSeries` forms; the trace-level engine is the Clebsch–Gordan law $S_k S_m = \sum_{j \leq m} S_{k+m-2j}$, `clebsch_gordan_trace` in `RSLocalCauchy`.)

Proposition 2.3 (local positivity, exact; machine-checked). *Let rs_k denote the k -th coefficient of the local Rankin–Selberg factor $((1-\alpha^2 x)(1-x)^2(1-\alpha^{-2}x))^{-1}$. Then $rs_{k+2} - rs_k = S_{k+2}(\alpha)^2$ with $rs_0 = S_0^2$, $rs_1 = S_1^2$; hence $rs_k = \sum_{j \leq k/2} S_{k-2j}(\alpha)^2$, and at a unit strand every rs_k is a nonnegative real — positivity is an identity, not an estimate.*

(`rs_local_cauchy`, `rsCoeff_eq_sum_sq`, `rs_local_positivity`.) The record stops at nine because the Langlands–Shahidi list of Eisenstein configurations is exhausted at E_8 [6]; no Eisenstein series in existence reaches $r \geq 10$.

3 Twisted families, grade, and the joint moment matrix

The following automorphic *vehicle* identities carry analytic control past the naive pair list. All decompositions below are machine-checked as exact identities of coefficient banks; the analytic conclusions use the cited classical inputs: automorphy of the Kim–Shahidi product $\sigma_2 \boxtimes \sigma_3$ in both roles [3], of $\text{Sym}^{\leq 4}$ [1, 3, 4], Rankin–Selberg theory for automorphic pairs [2], and nonvanishing on $\text{Re } s = 1$ [9].

Theorem 3.1 (twisted families at Serre grade). *For every $\mathrm{GL}(2)$ cusp form g and every $\mathrm{GL}(3)$ cusp form τ :*

$$\begin{aligned} L((g \boxtimes \mathrm{Sym}^2 \pi) \times \mathrm{Sym}^3 \pi, s) &= L(g \times \mathrm{Sym}^5 \pi, s) L(g \times \mathrm{Sym}^3 \pi, s) L(g \times \pi, s), \\ L((g \boxtimes \mathrm{Sym}^2 \pi) \times \mathrm{Sym}^4 \pi, s) &= L(g \times \mathrm{Sym}^6 \pi, s) L(g \times \mathrm{Sym}^4 \pi, s) L(g \times \mathrm{Sym}^2 \pi, s), \\ L(\mathrm{Sym}^4 \pi \times (\pi \boxtimes \tau), s) &= L(\mathrm{Sym}^5 \pi \times \tau, s) L(\mathrm{Sym}^3 \pi \times \tau, s). \end{aligned}$$

In each identity the left side and all right-hand factors except one are Rankin–Selberg L -functions of pairs of automorphic representations; hence $L(g \times \mathrm{Sym}^5 \pi, s)$, $L(g \times \mathrm{Sym}^6 \pi, s)$ and $L(\mathrm{Sym}^5 \pi \times \tau, s)$ are holomorphic and nonvanishing on $\mathrm{Re} s \geq 1$.

(Bank identities: `quinticG_bank`, `vehicle_bank`, `quinticTau_bank` — the last universal over an arbitrary abstract $\mathrm{GL}(3)$ Satake family, so no automorphy enters the compiled layer.)

Proposition 3.2 (grade dichotomy). *The division chains of Theorem 3.1 yield holomorphy and nonvanishing on the closed half-plane $\mathrm{Re} s \geq 1$ (Serre grade) and cannot yield entirety on $\mathbb{C}$ (converse grade): a quotient inherits poles at the interior zeros of its denominator, and no unconditional zero-free region covers the critical strip. Converse-theorem applications therefore require entirety inputs of the Kim–Shahidi type [7], which division does not supply.*

Corollary 3.3 (joint moment matrix for a Maass pair). *Let π, π' be Hecke–Maass cusp forms, not twist-equivalent. The mixed moments $\frac{1}{\pi(N)} \sum_{p \leq N} S_a(\theta_p(\pi)) S_b(\theta_p(\pi'))$ converge to 0 for all (a, b) with $a, b \leq 4$, for $(a, b) = (r, 0)$ with $r \leq 9$, and for*

$$(a, b) \in \{(5, 1), (6, 1), (5, 2)\},$$

the last three via Theorem 3.1.

The compiled layer identifies the joint-moment averages with the prime averages of the corresponding banks: the degree-one local Euler coefficient is the sum of the local weights (`radialLocalEulerCoeff_one`), hence any bank’s value at a prime is its local weight sum (`bank_at_prime`), hence the three banks read $S_5 S_1, S_6 S_1, S_5 S_2$ at every prime (`joint_51_at_prime` etc., file `JointMoments`).

Remark 3.4 (record check). Whether the entries $(5, 1), (6, 1), (5, 2)$ — and the $\mathrm{GL}(3)$ -twisted family of Theorem 3.1 — appear in the existing joint-Sato–Tate literature is an open bibliographic obligation of this note; a targeted search of the Kim–Shahidi applications papers and recent joint-distribution work did not surface them, and no novelty is claimed pending a full reading.

4 Structural insufficiency theorems

Any future proof of the conjecture must contain content of a specific shape. Two theorems delimit it.

Theorem 4.1 (Satake-blind payloads are insufficient). *No family of completed analytic objects that is independent of the Satake data can supply the twisted inputs of any converse theorem. In particular: the carrier completion attached to the polynomial Satake class satisfies identical functional equations for distinct data and distinct twists (machine-checked: the*

completed transform of the radius-live niceness class is literally the same function for different base pairs and twists), while the one-sided converse inputs distinguish real objects — the unit (zeta) datum inhabits the class and its one-sided reflection is refuted (`zeta_no_package`, `zeta_reflection_fails`), and the maximally non-tempered spiral datum inhabits the radius-live class while a $\mathrm{GL}(2)$ automorphic representation with its parameters would contradict Rankin–Selberg bounds. A discrimination-free source cannot feed a discrimination device.

Theorem 4.2 (the monoid gap at $m = 4$). *Let $\mathcal{A}$ be the monoid of automorphic objects generated from $\mathrm{GL}(2)$ cusp forms by $\mathrm{Sym}^{\leq 4}$, the products $\mathrm{GL}(2) \boxtimes \mathrm{GL}(2)$ [8] and $\mathrm{GL}(2) \boxtimes \mathrm{GL}(3)$ [3], isobaric sums, duals and character twists. Then for a $\mathrm{GL}(4)$ cusp form τ_4 , every factorization of $\tau_4 \otimes \mathrm{Sym}^5 \pi$ as $A \otimes B$ with $A, B \in \mathcal{A}$ requires an automorphic product on $\mathrm{GL}(2) \times \mathrm{GL}(4)$ or $\mathrm{GL}(3) \times \mathrm{GL}(3)$ — neither of which exists [11]. The verification is finite bookkeeping over the rank splits, including the self-dual sub-families $\tau_4 = g \boxtimes h$ and $\tau_4 = \mathrm{Sym}^3 h$.*

Together with the exhaustion of the Langlands–Shahidi list at E_8 and the failure of Euler structure for four or more copies of a $\mathrm{GL}(2)$ form against Siegel–Eisenstein pullbacks (the trilinear invariant functional is unique; from four factors on, invariant functionals smear into spectral integrals whose inversion presupposes the sought continuation), these results close every known route to rung ten and beyond.

5 The frontier, exactly

Proposition 5.1 (two faces of the gateway). *$\mathrm{Sym}^5 \pi$ automorphy follows from either: (A) niceness of the twist family $L(\tau_4 \times \mathrm{Sym}^5 \pi, s)$ at converse grade — by Theorem 4.2 requiring one new functorial product; or (B) the global converse theorem for $\mathrm{GL}(6)$ with twists up to $\mathrm{GL}(3)$ (the global Jacquet conjecture at $n = 6$, open; its local analogue is a theorem [12, 13]) together with converse-grade entirety of the $m \leq 3$ family, which by Proposition 3.2 the present division chains do not supply.*

Even both faces falling extends the tower only finitely (to Sym^6 , with the compiled cascade identity $(\mathrm{std}_\pi \otimes \tau) \otimes \mathrm{Sym}_\pi^5 = (\mathrm{Sym}_\pi^6 \oplus \mathrm{Sym}_\pi^4) \otimes \tau$, `sexticTau_bank`, and the quintic square $\mathrm{Sym}^5 \otimes \mathrm{Sym}^5 = \bigoplus_{j \leq 5} \mathrm{Sym}^{10-2j}$, `quinticSquare_bank`, ready as pure citation): the twist demand of each successive converse grows linearly while the product supply is fixed, so the full conjecture requires an engine not present in current mathematics — the Galois-theoretic engine is unavailable for Maass forms, and the trace-formula program remains unexecuted beyond its first cases. The open input of this entire subject is therefore *one object*: a coefficient-dependent mechanism producing the one-sided twisted reflections at unbounded rank. Its minimal test case is the $(r, m) = (5, 4)$ reflection.

Remark 5.2 (falsifiability). The reduction of Theorem 1.1 is falsifiable at every stage: a computed violation of the nine-moment convergence, of local positivity (Proposition 2.3), or of any bank identity of Proposition 2.2 at any n would refute the corresponding machine-checked statement. No violation is known; the identities have been verified symbolically at every n by compilation.

6 What is achieved and what is not

Achieved, unconditionally: the exact zero-slack reduction; the nine-moment single-form record with its coefficient algebra machine-checked end to end; exact local Rankin–Selberg positivity;

the three new joint-moment entries at Serre grade (subject to Remark 3.4); the two insufficiency theorems; and the reduction of the frontier to the two named faces of Proposition 5.1.

Not achieved, by anyone: the Sato–Tate conjecture for a single Maass form. Its remaining content is exactly the object named at the end of Section 5, and no assembly of existing results closes it — that negative statement is itself proved here at the three levels of Theorems 4.1 and 4.2 and the exhaustion facts of Section 4.

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